# File Handling in Python – Complete Detailed Notes

## 1. What is File Handling?

File Handling in Python means working with files to **store, read, update, and manage data permanently**. Unlike variables, file data remains saved even after the program stops. Python provides built-in functions to open, read, write, append, and close files. (GeeksforGeeks)

### Real-Life Examples:

- Saving student records
- Storing login credentials
- Creating reports
- Reading configuration files
- Managing logs

## 2. Why Use File Handling?

Without files:

- Data is temporary (lost after program ends)

With files:

- Data stays saved permanently
- Large data can be managed efficiently
- Programs can exchange data using files

## 3. Types of Files in Python

### A. Text Files

Stores readable text.

Examples:

- .txt
- .csv
- .json
- .py

### B. Binary Files

Stores data in bytes.

Examples:

- .jpg
- .png
- .mp3
- .pdf
- .exe

(GeeksforGeeks)

## 4. Opening a File

Use open() function.

### Syntax:

file = open("filename.txt", "mode")

### Example:

file = open("data.txt", "r")

- First argument = file name/path
- Second argument = mode

(GeeksforGeeks)

## 5. File Modes in Python

| Mode | Meaning |
|---|---|
| r | Read only |
| w | Write (overwrite/create) |
| a | Append |
| x | Create new file |
| r+ | Read + Write |
| w+ | Write + Read |
| a+ | Append + Read |
| b | Binary mode |
| t | Text mode (default) |

(GeeksforGeeks)

## 6. Reading a File

### A. read()

Reads complete file.

```
file = open("data.txt", "r")
content = file.read()
print(content)
file.close()
```

### B. read(n)

Reads first n characters.

```
file = open("data.txt", "r")
print(file.read(5))
file.close()
```

### C. readline()

Reads one line.

```
file = open("data.txt", "r")
print(file.readline())
file.close()
```

### D. readlines()

Reads all lines into list.

```
file = open("data.txt", "r")
print(file.readlines())
file.close()
```

(GeeksforGeeks)

## 7. Writing to File

### write()

```
file = open("data.txt", "w")
file.write("Hello Python")
file.close()
```

⚠ w mode erases old content first.

(programiz.com)

## 8. Appending Data

Adds content at end.

```
file = open("data.txt", "a")
file.write("\nNew line added")
file.close()
```

## 9. Creating a New File

Use x mode.

```
file = open("newfile.txt", "x")
file.close()
```

If file exists → error.

## 10. Closing a File

Use:

```
file.close()
```

Why important:

- Saves data properly
- Frees memory/resources

## 11. Best Method – with open()

Automatically closes file.

```
with open("data.txt", "r") as file:
    print(file.read())
```

Best practice in Python. (programiz.com)

## 12. File Pointer Functions

### tell()

Current cursor position.

```
file = open("data.txt", "r")
print(file.tell())
```

### seek(position)

Move cursor.

```
file.seek(0)
```

## 13. Example of seek()

```
file = open("data.txt", "r")
print(file.read(5))
file.seek(0)
print(file.read(5))
file.close()
```

## 14. File Exists Check

```
import os
```

```
if os.path.exists("data.txt"):
    print("Exists")
else:
    print("Not found")
```

## 15. Delete File

```
import os
os.remove("data.txt")
```

## 16. Delete Folder

```
os.rmdir("folder")
```

## 17. Exception Handling in Files

```
try:
    file = open("abc.txt", "r")
    print(file.read())

except FileNotFoundError:
    print("File not found")
```

(DigitalOcean)

## 18. Binary File Example

```
file = open("image.jpg", "rb")
data = file.read()
```

```
file.close()
```

## 19. Copy File Example

```
with open("a.txt", "r") as f1:
    data = f1.read()

with open("b.txt", "w") as f2:
    f2.write(data)
```

## 20. Important Methods Summary

| Method | Work |
|---|---|
| read() | Read all |
| readline() | Read one line |
| readlines() | Read all lines list |
| write() | Write data |
| writelines() | Write many lines |
| close() | Close file |
| tell() | Current position |
| seek() | Move cursor |

## 21. Real World Uses

- Attendance system
- Banking transactions
- Reports generation

- Student records
- Inventory systems
- Logs

## 22. Common Errors

| Error | Meaning |
|---|---|
| FileNotFoundError | File missing |
| PermissionError | No permission |
| IsADirectoryError | Path is folder |
| UnicodeDecodeError | Wrong encoding |

## 23. Best Practices

✅ Use with open() ✅ Close files properly ✅ Use try-except ✅ Use correct mode ✅ Backup before overwrite

❌ Avoid:

```
open("data.txt","w")
```

without checking important content.

## 24. Short Notes for Exam

- open() opens file
- read() reads file
- write() writes data
- append() adds data
- close() closes file
- with open() auto closes file

# 25. Mini Programs

## Program 1 – Write Name

```
with open("name.txt", "w") as file:
    file.write("Aadipal")
```

## Program 2 – Read File

```
with open("name.txt", "r") as file:
    print(file.read())
```

## Program 3 – Append

```
with open("name.txt", "a") as file:
    file.write("\nPython")
```

# 26. Interview Questions

## Q1. Difference between w and a?

- w overwrites file
- a adds at end

## Q2. Best way to open file?

Answer: with open()

## Q3. Which mode creates new file only?

Answer: x

## 27. Memory Trick

**ORWAC**

- **O**pen
- **R**ead
- **W**rite
- **A**ppend
- **C**lose

## 28. One-Line Summary

File Handling in Python is used to store and manage data permanently using files.

aq